

Chapter 2: Basic Concepts in Argument

Introduction

Everyone thinks. Everyone reasons. Everyone argues. And everyone is subjected to the reasoning and arguing of others. We are bombarded daily with reasoning from many sources: books, speeches, radio, TV, newspapers, employers, friends, and family.

Some people think well, reason well, and argue well. Some do not. The ability to think, reason, and argue well is partly a matter of natural gifts. But whatever our natural gifts, they can be refined and sharpened. **And the study of logic is one of the best ways to refine one's natural ability to reason and argue.**

Through the study of logic, one learns strategies for thinking well, common errors in reasoning to avoid, and effective techniques for evaluating arguments.

What is Logic?

But what is logic? Roughly speaking, logic is the study of methods for evaluating arguments. More precisely, logic is the study of methods for evaluating whether the premises of an argument adequately support (or provide good evidence for) its conclusion.

To better grasp what logic is, we need to understand the key concepts involved in this definition: argument, conclusion, premise, and support.

WHAT IS AN ARGUMENT?

Arguments are composed of one or more premises and a conclusion.

- **Premises** are statements in an argument offered as evidence or reasons why we should accept another statement, the conclusion.

- **Conclusion** is the statement in an argument that the premises are intended to prove or support.

So, an argument is a group of statements, one or more of which (called the premises) are intended to prove or support another statement (called the conclusion).

A **statement** is a sentence that can be viewed as either true or false.

Five things should be noted about statements

First, a sentence may be used to express more than one statement. For example, the grammatical sentence *Roses are red and violets are blue*.

Second, a statement can sometimes be expressed as a phrase or an incomplete clause, rather than as a complete declarative sentence. Consider the sentence:

”With mortgage interest rates at thirty-year lows, you owe it to yourself to consider refinancing your home.” (radio ad)

- This sentence, despite lacking a complete declarative structure, conveys a statement urging the listener to consider refinancing their home.
- The main statement is expressed in the form of an imperative phrase, ”you owe it to yourself to consider refinancing your home.”
- This example demonstrates how statements can be expressed in various forms, including phrases or incomplete clauses, while still conveying a clear message or proposition.

Five things about statements (Cont.)

Third, not all sentences are statements. For example, questions, proposals, exclamations, and suggestions are not statements.

Fourth, statements can be subjective matters of personal experience and objectively verifiable matters of fact:

- **Subjective Statement:** ”I believe that chocolate ice cream is the most delicious flavor.”
 - This statement expresses a subjective opinion about personal taste preferences. It represents a subjective matter of personal experience.
- **Objective Statement:** ”The Earth revolves around the Sun.”

- This statement represents an objectively verifiable matter of fact. It is based on scientific observations, evidence, and the heliocentric model of the solar system.

Five things about statements (Cont.)

Fifth, Some sentences that look like nonstatements are actually statements and can be used in arguments.

Example: "Alyssa, you should quit smoking. Don't you realize how bad that is for your health?"

- Although this sentence appears to be a suggestion or command, it can be understood as a statement when used in an argument. The speaker is presenting a premise or claim, asserting that Alyssa should quit smoking because it is detrimental to her health.
- The sentence functions as an argument by presenting a reason and appealing to the consequences of smoking on Alyssa's health.
- In general, sentences that resemble commands or suggestions can often be transformed into statements by adding a premise or claim and providing reasoning or evidence to support the intended argument.

IDENTIFYING PREMISES AND CONCLUSIONS

- In identifying premises and conclusions, we are often helped by **indicator words**.
- **Indicator words** are words or phrases that provide clues that premises or conclusions are being put forward.
- **Premise indicators** indicate that premises are being offered.
- **Conclusion indicators** indicate that conclusions are being offered.

Premise Indicators

- Since
- because
- for

- given that
- seeing that
- considering that
- inasmuch as
- as
- in view of the fact that
- as indicated by
- judging from
- on account of

Conclusion Indicators

- Therefore
- thus
- Hence
- consequently
- So
- accordingly
- it follows that
- for this reason
- that is why
- which shows that
- wherefore
- this implies that
- as a result

- this suggests that
- this being so
- we may infer that

A Note on Indicators

Note: the existence of indicators can never be a guarantee for a sentence to be premise or conclusion of an argument. Consider the following examples:

- I haven't seen you *since* high school.
- You've had that jacket *for* as long as I've known you.
- *Thus* far everything has been great.
- It was *so* cold that even the ski resorts shut down.
- I wouldn't mind *seeing that* movie again.
- There is water on the floor *because* the sink overflowed.

it's so important to consider the context when determining the meaning of an expression.

Arguments without Indicator Words

Sometimes arguments may contain no indicator words. Consider the example below:

"Achieving financial success is not solely determined by individual effort. There are systemic inequalities, economic downturns, and unpredictable market forces that can hinder one's ability to succeed. Additionally, personal circumstances such as family background, access to education, and social networks also play a significant role."

- In this example, the speaker is making an argument that highlights external factors and systemic influences that can impact an individual's ability to achieve financial success.
- It's important to note that arguments without explicit indicator words can still convey a perspective. These arguments highlight the complex interplay between individual agency and external influences in various aspects of life.

Finding the Conclusion without Indicators

How can we find the conclusion of an argument when the argument contains no indicator words?

- Find the main issue and ask yourself what position the writer or speaker is taking on that issue.
- Look at the beginning or end of the passage; the conclusion is often (but not always) found in one of those places.
- Ask yourself, "What is the writer or speaker trying to prove?" That will be the conclusion.
- Try putting the word **therefore** before one of the statements. If it fits, that statement is probably the conclusion.
- Try the "because" trick.

WHAT IS NOT AN ARGUMENT?

- To understand arguments thoroughly, it is crucial to understand what an argument is not.
- Types of non-argumentative discourse that are sometimes confused with arguments:
 - Reports
 - unsupported assertions
 - Loosely associated statements
 - Expository passages
 - conditional statements
 - Illustrations
 - explanations

Reports

A report is a set of statements intended to provide information about a situation, topic, or event. A report may contain many informational statements without containing any arguments. For instance:

Example: A wildlife conservation report states that the population of endangered tigers has declined by 95% over the past century due to habitat loss and illegal poaching. Furthermore, it emphasizes the importance of implementing strict conservation measures and raising awareness to protect the remaining tiger population.

This passage represents a report as it presents a set of informational statements... It provides factual information... While the report contains recommendations and calls to action, it does not feature explicit arguments with premises and conclusions. Instead, it primarily aims to inform the reader...

Unsupported Assertions

- Unsupported assertions are statements about what a speaker or writer happens to believe. Such statements can be true or false, rational or irrational, but they are parts of arguments only if the speaker or writer claims that they follow from, or support, other claims. Here is an example:

I believe that it is not dying that people are afraid of. Something else, something more unsettling and more tragic than dying frightens us. We are afraid of never having lived, of coming to the end of our days with the sense that we were never really alive, that we never figured out what life was for.

- This passage consists of a series of unsupported assertions, as the speaker expresses personal beliefs without providing explicit arguments or supporting evidence.
- While these assertions convey the speaker's perspective, they do not present logical reasoning, premises, or evidence to support the claims. Therefore, they remain unsupported assertions rather than forming part of a structured argument.

Loosely Associated Statements

- Loosely associated statements refer to a collection of statements that are loosely connected in terms of their topic or subject matter. These statements may share a common theme or have some superficial connection, but they lack a clear logical or argumentative structure.
- Loosely associated statements often involve expressing personal opinions, sharing unrelated experiences or observations, or presenting ideas without a coherent line of reasoning.

- These statements may be used in casual conversations, storytelling, or expressing personal thoughts and feelings. However, they do not form a cohesive argument or provide a structured line of reasoning.

Loosely Associated Statements: Example

Example:

Statement 1: I enjoy hiking in the mountains. The fresh air and beautiful views are invigorating.

Statement 2: My favorite color is blue. It reminds me of the ocean and brings a sense of calm.

Statement 3: Yesterday, I tried a new recipe. It was a spicy curry dish that had a unique blend of flavors.

- In this example, the statements are loosely associated as they touch upon different topics without forming a coherent argument.
- While these statements may share a general theme of personal preferences or experiences, they do not present a logical progression or provide premises to support a conclusion. Each statement stands alone.

Expository passage

Consider the example:

”The Industrial Revolution, which took place from the late 18th to the early 19th century, brought about significant changes in society. It marked a shift from agrarian and manual labor-based economies to industrialized and machine-driven production... The Industrial Revolution had profound social, economic, and environmental impacts, transforming the way people lived and worked.”

- This passage provides an exposition of the Industrial Revolution, presenting factual information about its defining features, technological advancements, and societal consequences. It aims to add upon and expand the topic sentence.
- The passage does not present a specific claim or assert a particular viewpoint that needs to be supported by evidence or logical reasoning. The purpose is to provide a clear and coherent explanation.

Conditional Statements

- A conditional statement is an *if-then* statement. Here are examples:
 - If it rains, then the picnic will be canceled.
 - You must speak French if you grew up in Quebec.
 - If at first you don't succeed, don't try skydiving.
- Conditional statements are made up of two basic parts. The part following *if* is the **antecedent**. The part following *then* is the **consequent**.
- Conditional statements need not be explicitly in *if-then* form. For example:
 - Should it rain, the picnic will be canceled.
 - In the event of rain, the picnic will be canceled.
 - Peter will graduate, provided he passes Critical Thinking.

Conditional Statements and Arguments

The relation between conditional statements and arguments may now be summarized as follows:

- A single conditional statement is not an argument.
- A conditional statement may serve as either the premise or the conclusion (or both) of an argument.
- The inferential content of a conditional statement may be re-expressed to form an argument.

Note: a conditional statement can be re-expressed to form an argument if and only if the antecedent of the conditional is only a necessary condition for its consequent.

Conditional Statements: Examples

1. A single conditional statement is not an argument:

Example: "If it rains, I will bring an umbrella."

This sentence is a single conditional statement... on its own, it does not present a complete argument.

2. **A conditional statement may serve as either the premise or the conclusion... of an argument:**

Premise: "If you study consistently, you will perform well on the exam."

Conclusion: "Therefore, you should study consistently."

In this example, the conditional statement serves as the premise of the argument.

3. **The inferential content of a conditional statement may be re-expressed to form an argument:**

Conditional Statement: "If you exercise regularly, you will improve your fitness."

Re-expressed Argument: "Since regular exercise leads to improved fitness, it is advisable to incorporate exercise into your routine."

In this example, the original conditional statement is re-expressed to form an argument.

Necessary and sufficient conditions

- A condition A is said to be **necessary** for a condition B, if (and only if) the falsity of A guarantees the falsity of B.
- A condition A is said to be **sufficient** for a condition B, if (and only if) the truth of A guarantees the truth of B.
- The relationship between the two conditions must be exactly one of the following four possibilities:
 1. The first is a necessary, but not a sufficient, condition for the second; or
 2. The first is a sufficient, but not a necessary, condition for the second; or
 3. The first is both a necessary and a sufficient condition for the second; or
 4. The first is neither a necessary nor a sufficient condition for the second.

Examples of Conditions

The relationship between the two conditions can have four possibilities:

- **a) The antecedent is a necessary, but not a sufficient, condition for the consequent:**

Example: "Having a driver's license is necessary, but not sufficient, for operating a commercial vehicle."

- **b) The antecedent is a sufficient, but not a necessary, condition for the consequent:**

Example: "Owning a smartphone is sufficient, but not necessary, for accessing the internet."

- **c) The antecedent is both a necessary and a sufficient condition for the consequent:**

Example: "Today's being neither Saturday nor Sunday is both a necessary and a sufficient condition for today's being a weekday."

- **d) The antecedent is neither a necessary nor a sufficient condition for the consequent:**

Example: "Wearing sunglasses is neither a necessary nor a sufficient condition for rain."

Exercise

Identify which conditions best fits in the following relations

1. Being over 18 years old is _____ condition to vote in an election.
2. Having a valid driver's license is _____ condition for legally driving a car.
3. Wearing a hat is _____ condition for it to rain.
4. Passing all required courses is _____ condition for graduation.
5. Having a valid passport is _____ condition for traveling internationally.
6. Owning a guitar is _____ condition for being a musician.
7. Eating a balanced diet is _____ condition for maintaining good health.

Illustrations

Illustration is the use of examples to make ideas more concrete and to make generalizations more specific and detailed.

Claim: Regular exercise contributes to overall physical well-being.

Illustration: For instance, engaging in aerobic exercises such as jogging, cycling, or swimming can improve cardiovascular health... Furthermore, incorporating strength training exercises, such as weightlifting or resistance training, helps build muscle strength and bone density...

In this example, the claim is that regular exercise contributes to overall physical well-being. To make this claim more concrete and specific, the illustration provides examples of specific exercises and explains how they impact various aspects of physical health.

Explanations

- Consider the following two statements:
 - Capital punishment should be abolished *because* innocent people may be mistakenly executed.
 - Titanic sank *because* it struck an iceberg.
- In the first example, the clause that precedes the word "because" is what the author is trying to convince us of. In the second, the clause that precedes "because" is what the author wants to explain. One indicator of the difference is that "Capital punishment should be abolished" is an interesting and controversial claim. It makes sense that someone would try to argue for it; whereas "Titanic sank" is an accepted fact that no one would need to argue for.

Explanations and Arguments

- Sometimes we will have different explanations for why something is the case. If we are engaging in opting for which explanation outweighs the other, we are making an argument.
- Look at the following example:

Statement: The car won't start.

Explanation 1: The car won't start because the battery is dead.

Explanation 2: The car won't start because there is a problem with the ignition system.

Argument: The car won't start, and the explanation that outweighs the other is that the battery is dead.

- In this example, there are two different explanations... The argument is made to determine which explanation carries more weight.
- To support the argument, additional evidence or reason may be presented.

Explanation as Conclusion

- Sometimes an explanation can itself be the conclusion of an argument.
- Consider the example:

Premise 1: The ground is wet.

Premise 2: It has been raining heavily.

Conclusion (Explanation): Therefore, the wet ground is a result of the heavy rainfall.

- In this example, the conclusion itself serves as an explanation. The premises establish that the ground is wet and that there has been heavy rainfall. Based on these premises, the conclusion explains the relationship between the wet ground and the heavy rainfall...

Distinguishing Arguments from Explanations

- An **explanation** tries to show **why** something is the case, not to prove **that** it is the case.
- Explanations have two parts. The statement that is explained is the **explanandum**. The statement that does the explaining is the **explanans**.
- In everyday speech, we often use "argument" and "explanation" almost interchangeably. How then does one distinguish arguments from explanations? There are **four basic tests**.

The Four Tests (I)

The Common-Knowledge Test

Is the statement that the passage seeks to prove or explain a matter of common knowledge? If it is, the passage is probably an explanation rather than an argument. (There's usually little point in trying to prove something that is already a well-known fact.)

The Past-Event Test

Is the statement that the passage is seeking to prove or explain an event that occurred in the past? If so, the passage is probably an explanation rather than an argument because it is much more common to try to explain *why* past events have occurred rather than to prove *that* they occurred.

The Four Tests (II)

The Author's Intent Test

Is it the speaker's or writer's intent to prove or establish *that* something is the case—that is, to provide reasons or evidence for accepting a claim as true? Or is it his intent to explain *why* something is the case...? If the former, the passage is an argument; if the latter, the passage is an explanation.

The Principle of Charity Test

The principle of charity... requires that we always interpret unclear passages generously and, in particular, that we never interpret a passage as a bad argument when the evidence reasonably permits us to interpret it as not an argument at all. This test often proves helpful when the other tests yield no clear answer.

DEDUCTION AND INDUCTION

- Before we can effectively evaluate an argument, we need to understand what kind of argument is being offered. Traditionally, arguments have been divided into two types: **deductive arguments** and **inductive arguments**.
- It is often said that the difference between deduction and induction is that *deduction* moves from general premises to particular conclusions, whereas *induction* moves from particular premises to general conclusions. That, however, is a misconception.

Misconception Example 1

Example of a deductive argument that moves from particular premises to a general conclusion:

- **Particular Premise 1:** Every time I eat peanuts, I have an allergic reaction.
- **Particular Premise 2:** Every time my friend eats peanuts, they have an allergic reaction.
- **Particular Premise 3:** Every time my cousin eats peanuts, they have an allergic reaction.
- **Conclusion:** Therefore, it can be concluded that people who are allergic to peanuts will have an allergic reaction when they consume peanuts.

In this example, the particular premises establish a consistent pattern. The deductive argument then draws a general conclusion... The argument relies on the principle of generalizing from specific instances to form a broader conclusion.

Misconception Example 2

Example Deductive argument that proceeds from general premises to a general conclusion

General Premise 1: The majority of people who exercise regularly have better physical health.

General Premise 2: The majority of people who eat a balanced diet have better physical health.

General Conclusion: Therefore, the majority of people who exercise regularly and eat a balanced diet have better physical health.

In this example, the general premises establish two statements based on observed patterns... The deductive argument then combines these general premises to draw a general conclusion... The conclusion is derived from the general trend observed in both premises.

Misconception Example 3

Example of deductive argument that proceeds from particular to particular

Example 1

Particular Premise 1: Every time I press the power button on my computer, it turns on.

Particular Premise 2: I just pressed the power button on my computer.

Particular Conclusion: Therefore, my computer has turned on.

Example 2

Specific Premise 1: All roses are flowers.

Specific Premise 2: This object is a rose.

Specific Conclusion: Therefore, it can be concluded that this object is a flower.

Misconception Example 4

Example of an inductive argument that moves from general premises to a particular conclusion:

General Premise 1: Studies have shown that regular exercise improves cardiovascular health.

General Premise 2: John engages in regular exercise.

General Premise 3: Mary engages in regular exercise.

Conclusion: Therefore, it is likely that both John and Mary have improved cardiovascular health.

In this example, the general premises establish a pattern or trend... The conclusion then applies this general pattern to a particular situation... However, it is important to note that the conclusion is a probable or likely inference, not a certain one...

Misconception Example 5

Example of an inductive argument that moves from particular premises to a particular conclusion:

Particular Premise 1: John ate a large portion of spicy food.

Particular Premise 2: John experienced heartburn after eating the spicy food.

Particular Conclusion: Therefore, it is likely that John's heartburn was caused by the spicy food he ate.

In this revised example, the particular premises establish a specific situation... The inductive argument then draws a particular conclusion, suggesting that the heartburn was likely caused by the spicy food. The conclusion is based on the observed connection between the specific instances...

Misconception Example 6

Example of an inductive argument that proceeds from general premises to a general conclusion:

General Premise 1: Studies have shown that regular exercise improves cardiovascular health.

General Premise 2: Studies have shown that a balanced diet contributes to overall well-being.

General Conclusion: Therefore, engaging in regular exercise and maintaining a balanced diet likely promotes good overall health.

In this example, the general premises establish the findings from studies... The inductive argument then combines these general premises to draw a general conclusion... The conclusion is based on the observed patterns and evidence from multiple studies.

Key Differences between Deductive and Inductive Arguments

Deductive Arguments • The conclusion is said to follow **necessarily** from the premises.

- Aim to provide **certainty** and establish truths. If premises are true and argument is valid, the conclusion must be true.
- Tend to move from general principles to specific conclusions.
- Strength lies in its **validity**.
- Commonly used in mathematics, logic, and formal systems.

Inductive Arguments • The conclusion is based on observed patterns or evidence. The reasoning moves mainly from specific instances to a general conclusion.

- Do not aim for certainty. They provide conclusions that are **probable** or **likely**.
- Tend to move from specific instances to general conclusions.
- Strength lies in the **quality and quantity of evidence**.
- Commonly used in scientific inquiries, research, and everyday reasoning.

Tests for Deductive vs. Inductive

There are three tests that greatly simplify the task of determining whether an argument should be regarded as deductive or inductive:

- **Indicator Word Test:** This test involves examining the presence of indicator words or phrases. Words like "necessarily," "conclusively" indicate deduction. Words like "probably," "likely" suggest induction.
- **Strict Necessity Test:** This test examines whether the truth of the premises guarantees the truth of the conclusion. In a deductive argument, there is a strict logical necessity. In an inductive argument, it is probable but not logically necessary.
- **Common Pattern Test:** This test involves looking for common patterns or structures in arguments. Deductive arguments often follow specific logical patterns (modus ponens, syllogisms). Inductive arguments may include generalizations, analogies, or statistical reasoning.

The Indicator Word Test

Just as we use indicator words to signal the assertion of premises or conclusions, we use indicator words to signal when our arguments are deductive or inductive.

Deduction indicator words: • certainly

- it logically follows that
- definitely
- it is logical to conclude that
- absolutely
- this logically implies that
- conclusively
- this entails that

Induction indicator words: • Probably

- one would expect that
- Likely
- it is a good bet that

- it is plausible to suppose that
- chances are that
- it is reasonable to assume
- that odds are that

The Strict Necessity Test

- All deductive arguments claim, explicitly or implicitly, that their conclusions follow necessarily from their premises.
- The strict necessity test can be stated as follows:
 - An argument's conclusion either follows with strict logical necessity from its premises or it does not.
 - If the argument's conclusion **does** follow with strict logical necessity from its premises, the argument should always be treated as deductive.
 - If the argument's conclusion **does not** follow with strict logical necessity from its premises, the argument should normally be treated as inductive.
- Consider the following arguments:

Alan is a father. Therefore, Alan is a male. (deductive, why?)

Jill is a six-year-old girl. Therefore, Jill cannot run a mile in one minute flat. (inductive why? It is possible to imagine a situation where Jill, as an exceptional individual, could run a mile in one minute flat.)

The Common Pattern Test/typical kinds of arguments

Typically inductive arguments: • Argument based on prediction

- Argument based on analogy
- An inductive generalization
- An argument from authority
- An argument based on signs
- A causal inference

Typically deductive arguments: • Argument based on definition

- Argument based on mathematics
- Syllogism

COMMON PATTERNS OF DEDUCTIVE REASONING

Five common patterns of deductive reasoning:

- hypothetical syllogism
- categorical syllogism
- argument by elimination/disjunctive syllogism
- argument based on mathematics
- argument from definition

Hypothetical Syllogism

A syllogism is a three-line argument that contains at least one hypothetical or conditional (i.e., if-then) premise.

Example:

Premise 1: If people have access to quality education, then they are more likely to secure higher-paying jobs.

Premise 2: The government has invested in improving access to quality education.

Conclusion: Therefore, people are more likely to secure higher-paying jobs.

- This argument is deductive because it follows a hypothetical syllogism structure, containing an "if-then" premise. The first premise establishes a conditional relationship...
- The second premise states that the government has taken steps to improve access... By combining these premises, it logically follows that people are more likely to secure higher-paying jobs.

Categorical Syllogism

Categorical syllogism is defined as a three-line argument in which each statement begins with the word **All**, **Some**, or **No**.

Example 1:

Premise 1: All oaks are trees.

Premise 2: All trees are plants.

Conclusion: Therefore, all oaks are plants.

Example 2:

Premise 1: Some Democrats are elected officials.

Premise 2: All elected officials are politicians.

Conclusion: Therefore, some Democrats are politicians.

Categorical syllogisms are a well-known form of rigorous logical reasoning... Because they adhere to specific rules and logical patterns, categorical syllogisms are typically treated as deductive arguments.

Disjunctive syllogism/Argument by Elimination

An argument by elimination seeks to logically rule out (eliminate) various possibilities until only a single possibility remains.

Example:

- **Premise 1:** The thief who stole the jewelry must be either Anna, Ben, or Carla.
- **Premise 2:** Anna has an airtight alibi and was out of town during the time of the theft.
- **Premise 3:** Ben was caught on TV footage at a different location during the time of the theft.
- **Conclusion:** Therefore, the thief who stole the jewelry must be Carla.

This argument is deductive because the conclusion necessarily follows from the premises through the process of elimination... By eliminating the possibilities of Anna and Ben as the thief, the only remaining option is Carla.

Argument Based on Mathematics

Mathematics is a model of logical, step-by-step reasoning. Mathematicians don't claim that their conclusions are merely likely or probable. They claim to prove their conclusions on the basis of precise mathematical concepts and reasoning.

Example:

Premise 1: Light travels at a rate of 186,000 miles per second.

Premise 2: The sun is more than 93 million miles distant from the earth.

Conclusion: Therefore, it takes more than eight minutes for the sun's light to reach the earth.

This argument is deductive because it relies on precise mathematical concepts and reasoning to reach the conclusion... If the premises are true and the mathematical calculation is accurate, the conclusion must also be true.

Argument from Definition

In an argument from the definition, the conclusion is presented as being "true by definition," that is, as following simply by definition from some keyword or phrase used in the argument.

1. Janelle is a cardiologist. Therefore, Janelle is a doctor.

- In this example, the term "cardiologist" is defined as a medical specialist... Since the definition of "cardiologist" falls under the broader definition of "doctor," it is deductively true that Janelle... is also a doctor.

2. Bertha is an aunt. It follows that she is a woman.

- In this example, the term "aunt" is defined as the sister of one's parent... Since the definition of "aunt" specifies a familial relationship that is exclusive to females, it can be deduced that Bertha... is a woman.

Arguments from the definition are always deductive because if the relevant definition is true, the conclusion derived from it will also be necessarily true.

COMMON PATTERNS OF INDUCTIVE REASONING**Six common patterns of inductive reasoning:**

- inductive generalization
- predictive argument

- argument from authority
- causal argument
- statistical argument
- argument from analogy

An inductive generalization

An inductive generalization is an argument in which a generalization is claimed to be probably true based on information about some members of a particular class.

Example 1:

Premise: All dinosaur bones so far discovered have been more than sixty-five million years old.

Conclusion: Therefore, probably all dinosaur bones are more than sixty-five million years old.

Example 2:

Premise 1: Six months ago I met a farmer from Iowa, and he was friendly.

Premise 2: Four months ago I met an insurance salesman from Iowa, and he was friendly.

Premise 3: Two months ago I met a dentist from Iowa, and she was friendly.

Conclusion: I guess most people from Iowa are friendly.

The strength of these inductive generalizations relies on several factors, such as the representativeness and reliability of the available sample.

Predictive Argument

The prediction argument attempts to make claims about future effects. Here are two examples:

It has rained in Addis Ababa every April since weather records have been kept.
Therefore, it will probably rain in Addis Ababa next April.

Most U.S. presidents have been tall.
Therefore, probably the next U.S. president will be tall.

- Because nothing in the future is absolutely certain, arguments containing predictions are usually inductive.

Deductive Predictions

Note that: predictions can be argued for deductively:

Premise 1: All mammals give birth to live young.

Premise 2: A platypus is a mammal. (Platypuses are semi-aquatic mammals)

Conclusion: Therefore, the platypus will give birth to live young.

- This argument is deductive because it follows a logical structure where the conclusion necessarily follows from the premises.
- In this case, the conclusion is not a prediction about a future event in the usual sense, but rather a deduction based on the established category and the general rule.
- It is important to note that deductive predictive arguments are relatively rare as they require a clear logical connection between the premises and the predicted outcome.

Argument from analogy

- is a type of inductive argument, whereby perceived similarities are used as a basis to infer some further similarity that has yet to be observed.
- **Structure:**
 - P and Q are similar with respect to properties a, b, and c.
 - P has been observed to have further property x.
 - Therefore, Q probably has property x also.
- **Strength of argument based on:**
 - The relevance of the known similarities to the similarity inferred in the conclusion.
 - The degree of relevant similarity (or dissimilarity) between the two objects.
 - The amount and variety of instances that form the basis of the analogy.

Argument from analogy: Example

Example:

Premise 1: Humans and chimpanzees share a high degree of genetic similarity and have similar anatomical structures.

Premise 2: Humans have the ability to use language for communication.

Conclusion: Therefore, it is probable that chimpanzees also have the ability to use language for communication.

- This argument is an example of an argument from analogy, which is an inductive form of reasoning.
- The strength of this argument depends on several factors, including the relevance of the known similarities (genetics, anatomy) to the inferred similarity (language usage).
- While an argument from analogy can provide insights and generate hypotheses, it does not guarantee the truth of the conclusion. It relies on perceived similarities and inference rather than deductive certainty.

Causal Argument

Premise: Every time I eat ice cream, I get a headache.

Conclusion: Therefore, eating ice cream causes me to get a headache.

- This argument is an example of an inductive causal argument because it draws a general conclusion about a cause-and-effect relationship based on specific instances or observations.
- The argument does not provide a deductively certain conclusion because it is based on observed patterns rather than a guaranteed logical relationship. It is possible that there could be other factors causing the headaches.

Deductive Causal Arguments

- It cannot be assumed, however, that causal arguments are always inductive. The following causal argument, for example, is clearly deductive:

Whenever iron is exposed to oxygen, it rusts.

This iron pipe has been exposed to oxygen.

Therefore, it will rust.

- This argument is deductive because it follows a logical structure where the conclusion is necessarily true if the premises are true. The premises establish a general principle... and a specific instance... Based on the defined cause-and-effect relationship, the conclusion logically follows.

Statistical Argument

- A statistical argument rests on statistical evidence—that is, evidence that some percentage of some group or class has some particular characteristic.
- Here's an example:

Premise: Studies show that 80% of people who exercise for at least 30 minutes a day have lower risks of heart disease.

Conclusion: Therefore, engaging in regular exercise for at least 30 minutes a day is likely to reduce the risk of heart disease.

- The argument is an example of a statistical argument because it relies on statistical evidence to support the conclusion.
- Statistical arguments are typically inductive because they involve drawing general conclusions based on statistical evidence. The conclusion is presented as probable rather than certain.

Argument from authority

- An argument from authority, also called an appeal to authority, or *argumentum ad verecundiam* is a form of argument in which the opinion of an authority on a topic is used as evidence to support an argument.
- **Structure:**

Person A claims that X is true.

Person A is an expert in the field concerning X.

Therefore, X should be believed.

Argument from authority: Example

Here's an example of an argument from authority:

Premise 1: Dr. Smith, a renowned scientist, claims that climate change is primarily caused by human activities.

Premise 2: Dr. Smith is an expert in the field of climate science with multiple peer-reviewed publications.

Conclusion: Therefore, we should believe that climate change is primarily caused by human activities.

- In this example, the argument relies on the authority and expertise of Dr. Smith in the field of climate science.
- The given argument is inductive because it is based on the generalization that if a person is an expert in a specific field (Person A), their claims on that topic (X) are likely to be true. However, inductive reasoning does not provide certainty or guarantee the truth of the conclusion.

EVALUATION OF ARGUMENTS

In the proper sense of the term, an argument cannot be evaluated as simply good/bad, right/wrong, correct/incorrect, or acceptable/unacceptable. Instead, we employ unique terminologies to evaluate arguments.

For deductive arguments, the following terminologies are used:

valid, invalid, sound, and unsound.

For inductive arguments, the following terminologies are used:

strong, weak, cogent, and uncogent.

Valid Deductive Argument

- A valid deductive argument is a type of reasoning where the logical structure of the argument **guarantees** that if the premises are true, the conclusion must also be true. In other words, a deductive argument is valid when the conclusion logically follows from the premises.
- **Validity is determined by the form or structure of the argument**, regardless of whether the premises or conclusion are actually true or false. Even if the premises

of a valid deductive argument are false, as long as the logical structure is valid, the conclusion must still be true if the premises were true.

Validity and Truth Value

Note: truth value is not important for identifying validity.

Consider the following argument forms:

All adlers are bobkins.
 All bobkins are crockers.
 Therefore, all adlers are crockers.

All A are B.
 All B are C.
 So, All A are C.
 valid

If P then Q
 P
 so, Q
 valid

All A are B.
 All C are B.
 So, All A are C.
 invalid

If P then Q
 Q
 So, P
 invalid

Validity Doesn't Require Truth Value!

Premises	Conclusion	Validity
T	T	(Can be Valid)
T	F	Invalid
F	T	(Can be Valid)
F	F	(Can be Valid)

Valid Arguments with False Premises

- Some valid arguments have **false premises and a false conclusion**. For example:

Premise 1: If dragons breathe fire, then they have wings. (False)

Premise 2: Dragons do not have wings. (False)

Conclusion: Therefore, dragons do not breathe fire. (False)

This argument is valid because the form "If P then Q, and not Q, then not P" is always true, regardless of the truth of P or Q.

- Some valid arguments have **false premises and a true conclusion**:

Premise 1: All cats have wings.

Premise 2: Fluffy is a cat.

Conclusion: Therefore, Fluffy has wings.

While the premises of this argument are false (cats do not have wings), the argument itself is still considered valid because the conclusion logically follows from the premises.

The One Impossible Combination

Valid argument with **true premises and a true conclusion**:

Premise 1: All mammals are warm-blooded.

Premise 2: Dogs are mammals.

Conclusion: Therefore, dogs are warm-blooded.

- In this example, both premises are true, and the conclusion logically follows from the premises.
- There is, however, one combination of truth or falsity that no valid argument can have:
No valid argument can have all true premises and a false conclusion.

Invalid Deductive Argument

A deductive argument in which the conclusion does not follow necessarily from the premises is said to be an **invalid deductive argument**. Here are four examples:

True premise true conclusion

All dogs are animals.
Lassie is an animal.
Therefore, Lassie is a dog.

True premise false conclusion

If I'm a monkey's uncle, then I'm a primate.
I'm not a monkey's uncle.
So, I'm not a primate.

False premise true conclusion

All pears are vegetables.
All fruits are vegetables.
Therefore, all pears are fruits.

False premise false conclusion

All dogs are cats.
All cats are whales.
Therefore, all whales are dogs.

Soundness

Sound argument = valid + all true premise

Unsound argument = if at least one of the above conditions is missed.

Strong Inductive Argument

- A strong inductive argument is a type of reasoning where the premises provide significant support for the conclusion, making it **probable or likely** to be true.

- In a strong inductive argument, if the premises are true, then it is highly probable that the conclusion is also true. The **strength of an inductive argument is based on the quality and quantity of evidence**.
- It's important to note that unlike deductive arguments, inductive arguments **do not aim to establish certainty**. Instead, they aim to provide substantial evidence, such that if the premises are true, the conclusion is more likely to be true.

Strength and Truth Value

NOTE THAT: LIKE VALIDITY, STRENGTH DOESN'T REQUIRE TRUTH VALUE.

Premises	Conclusion	Strength
T	prob. T	(Can be Strong)
T	prob. F	Weak
F	prob. T	(Can be Strong)
F	prob. F	(Can be Strong)

Strong Arguments with False Premises

Strong arguments with false premises and a probably false conclusion:

Premise 1: Every time I use a specific brand of antivirus software, my computer is protected from all types of malware.

Premise 2: I am using that specific brand of antivirus software.

Conclusion: Therefore, my computer is fully protected from all types of malware.

In this example, the premises are false because no antivirus software can guarantee complete protection... However, the argument is considered strong because there may be a belief or marketing claims that suggest the software provides comprehensive protection.

Strong arguments with false premises and a probably true conclusion:

Premise 1: Every time I study for at least 10 hours, I perform exceptionally well on exams.

Premise 2: I studied for at least 10 hours for today's exam.

Conclusion: Therefore, I will likely perform exceptionally well on today's exam.

In this example, the premises are false because studying for a specific duration does not guarantee exceptional performance... However, the argument is considered strong because there may be a correlation between extensive studying and improved exam results.

The One Impossible Inductive Combination

A strong inductive argument with **true premises and a probably true conclusion**:

Premise 1: Every time I have taken my car to the mechanic for regular maintenance, it has run smoothly afterward.

Premise 2: I took my car to the mechanic for regular maintenance.

Conclusion: Therefore, my car will probably run smoothly afterward.

In this argument, the premises are true because regular maintenance can improve the performance and reliability of a car. The conclusion... is likely to be true based on the past experiences of the car owner.

Note: As with valid deductive arguments, **no strong inductive argument can have true premises and a probably false conclusion.**

Weak Inductive Arguments

Weak Inductive Arguments, like invalid deductive arguments, can have any combination of truth or falsity in the premises and conclusion. Here are some examples:

True premise and probably true conclusion

Most U.S. presidents have been married.

Therefore, probably the next U.S. president will be a man.

True premise and probably false conclusion

Most U.S. presidents have been over fifty years old.

Therefore, probably the next U.S. president will be single.

False premise and probably true conclusion

Most U.S. presidents have been women.

Therefore, probably the next U.S. president will be married.

False premise and probably false conclusion

Most U.S. presidents have been less than 5 feet tall.

Therefore, probably the next U.S. president will be single.

Cogency

Cogent argument: an argument both inductively strong and has all true premises.

An uncogent argument: an inductive argument that is either weak or has at least one false premise.

References – Books

1. **"A Rulebook for Arguments"** by Anthony Weston: This book provides a concise and accessible guide to understanding and constructing arguments. It covers topics such as identifying premises and conclusions, evaluating evidence, and avoiding common fallacies.
2. **"Critical Thinking: A Student's Introduction"** by Gregory Bassham, William Irwin, and Henry Nardone: This comprehensive textbook offers an in-depth exploration of critical thinking skills, including argument analysis, logical reasoning, and evaluating arguments.
3. **"Understanding Arguments: An Introduction to Informal Logic"** by Walter Sinnott-Armstrong and Robert Fogelin: This book introduces students to the basic principles of argumentation and critical thinking.
4. **"The Elements of Reasoning"** by Ronald Munson and Andrew Black: This book focuses on the elements of reasoning and argumentation, providing practical tools for analyzing and evaluating arguments.
5. **"Logic: A Very Short Introduction"** by Graham Priest: This concise introduction to logic covers both formal and informal aspects of reasoning.

References – websites

1. **Purdue Online Writing Lab (OWL) - Argument:** The OWL website provides a comprehensive guide on argumentation... (Website: owl.purdue.edu)

2. **Stanford Encyclopedia of Philosophy - Deductive and Inductive Arguments:** This online encyclopedia offers in-depth articles on deductive and inductive reasoning... (Website: plato.stanford.edu)
3. **Critical Thinking Web - Argument Basics:** The Critical Thinking Web provides interactive tutorials and exercises on argument basics... (Website: www.criticalthinkingweb.com)
4. **Internet Encyclopedia of Philosophy - Informal Logic:** This online encyclopedia features articles on informal logic... (Website: www.iep.utm.edu)
5. **Philosophy Terms - Argument:** Philosophy Terms offers a concise definition and explanation of arguments... (Website: philosophyterms.com)
6. **Georgetown University Writing Center - Evaluating Arguments:** The Georgetown University Writing Center offers a guide on evaluating arguments... (Website: writingcenter.georgetown.edu)
7. **Philosophy Now - Argument Evaluation:** Philosophy Now provides articles and resources on argument evaluation... (Website: philosophynow.org)
8. **TED-Ed - Lessons on Critical Thinking and Argumentation:** TED-Ed offers a collection of animated lessons on critical thinking and argumentation... (Website: ed.ted.com)